GRADE 10 / Life Sciences / English

Cells: structure explains function

01 Understand

Learning goals: Identify cell structures • Relate an organelle to its function • Compare typical plant and animal cells

Cells are the basic structural and functional units of living organisms. A cell membrane controls movement of substances into and out of the cell.

The nucleus contains genetic material. Mitochondria are sites of aerobic cellular respiration; they transfer energy into usable chemical forms.

Typical plant cells have a cellulose cell wall and a large central vacuole. Photosynthetic plant cells contain chloroplasts; not every plant cell has chloroplasts.

Compare structures, not just outlines

Structure	Typical plant cell	Typical animal cell
Cell membrane	present	present
Cellulose cell wall	present	absent
Chloroplasts	in photosynthetic cells	absent
Mitochondria	present	present

02 Follow the reasoning

A micrograph shows a rigid outer layer and many chloroplasts. Explain the most likely cell type.

1. The rigid outer layer is consistent with a cell wall.

2. Chloroplasts support photosynthesis, indicating a photosynthetic plant cell.

3. Use the observed structures as evidence; a rectangular shape alone is not enough to identify a cell reliably.

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Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. Name the structure that controls movement across a cell boundary.

2. Explain why muscle cells need many mitochondria.

3. A drawing is 40 mm long and the actual cell is 0.1 mm. Calculate magnification.

04 Check and correct

1. The cell membrane.

2. Muscle contraction needs energy; mitochondria support aerobic respiration and ATP production.

3. Magnification = drawing size / actual size = 40/0.1 = 400 times.

Common mistake to avoid

Do not say mitochondria create energy. They transfer energy during cellular respiration.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: Life at the molecular, cellular and tissue level: cells

DBE CAPS, Grades 10–12: Life Sciences. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Siyavula Grade 10 Life Sciences Learner Book, chapter 3: cell structures (indexed PDF pages 79–81). Available in the signed-in resource search. Original publisher attribution retained.

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID lisc-10.